

Agentic Finance Reconciliation System

— Documentation

Document version: 1.0 **Type:** End-to-end bank-statement reconciliation automation

Stack: n8n · Gmail API · AutoCount Accounting API/SQL · Python · WhatsApp/Email alerts

Redacted reference version. All credentials, API keys, client names, bank account numbers, and monetary examples shown here are illustrative placeholders. The workflow logic and business rules are preserved for reference.

1. Objective

Automate the monthly bank-statement reconciliation that finance staff previously did by hand for multiple clients. The system ingests bank statements from a dedicated mailbox, matches each transaction against the client's accounting records in AutoCount, applies a set of business rules to resolve near-matches, and flags anything unmatched to staff for action — with a full audit trail retained at every step.

2. High-Level Flow

The end-to-end pipeline proceeds through the following stages:

- A scheduled trigger checks a dedicated Gmail inbox for new bank-statement emails (per client).
 - Statement attachments (PDF / CSV / Excel) are downloaded and parsed into a normalized transaction list.
 - Each parsed transaction is normalized: date, amount, currency, direction (debit / credit), and a cleaned description.
 - Accounting records for the same client and period are pulled from AutoCount (via API / SQL).
 - The matching engine pairs bank transactions against accounting entries using layered rules.
 - Matched pairs are marked reconciled; unmatched items are flagged as exceptions.
 - A reconciliation summary is written to the audit log (per client, per run).
 - Exceptions are pushed to staff via WhatsApp and/or email for manual review.
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3. Statement Parsing

Bank statements arrive in inconsistent formats across banks, so parsing is format-aware:

- **PDF statements** — text extraction with a per-bank layout profile that locates the transaction table and column positions.
- **CSV / Excel statements** — column mapping per bank (date column, description, debit, credit, balance).
- **Normalization** — dates are coerced to ISO format, amounts to a single decimal convention (thousands separators stripped), and each row is tagged debit or credit.

Rows that cannot be confidently parsed are set aside and reported rather than silently dropped.

4. Matching Engine & Business Rules

Matching runs in layers, from strictest to most permissive. A bank transaction is compared against candidate AutoCount entries within the same client and period.

Rule	Condition	Outcome
Exact match	Same date, same amount, same direction	Auto-reconciled
Date-tolerance	Same amount and direction, date within $\pm N$ days	Auto-reconciled (settlement lag)
Fee-adjusted	Amount differs by a known bank-fee amount / range	Reconciled, fee difference noted
FX-adjusted	Foreign-currency entry; amounts match after applying the recorded FX rate within tolerance	Reconciled, FX note attached
Transfer pairing	Inter-account transfers matched as a debit/credit pair	Reconciled as transfer
No match	No candidate satisfies any rule	Flagged as exception

Each applied rule is recorded against the transaction so a reviewer can see *why* something reconciled (or didn't).

5. Exception Handling & Alerts

Anything the engine can't reconcile becomes an exception:

- Unmatched bank transactions (money moved, no accounting entry).
- Unmatched accounting entries (entry exists, no bank movement).
- Ambiguous matches (more than one candidate within tolerance).

Exceptions are batched per client and sent to the responsible staff member via WhatsApp and/or email, with enough context (date, amount, description, suspected reason) to action them quickly. Staff resolve exceptions in the source systems; the next run picks up the corrections.

6. Audit-Compliant Logging

Every run writes an immutable record capturing: client, period, run timestamp, counts of matched / unmatched / exception items, the rule applied to each matched transaction, and the raw parsed statement snapshot. This gives finance a defensible trail showing exactly what was reconciled, how, and when — important for audit and for investigating disputes after the fact.

7. Operational Outcomes

- Removes hours of manual line-by-line matching each month, per client.
 - Consistent rule application (date tolerance, fees, FX, transfers) instead of ad-hoc judgement.
 - Faster exception surfacing — staff see only the items that actually need a human.
 - A complete audit trail retained automatically.
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8. Access & Credentials

Gmail, AutoCount, and messaging-provider credentials are managed in the automation platform's credential store and are not included in this document. Contact the system administrator for access.